Genetic Ancestry and Socioeconomic Deprivation Interact to Influence Type 2 Diabetes Risk in the *All of Us* Cohort

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OBJECTIVE

- Determine whether there is evidence for disparities in type 2 diabetes (T2D) burden across race, genetic ancestry, and socioeconomic status in *All of Us*
- Assess how these variables interact to affect T2D risk

THE ALL OF US RESEARCH PROGRAM

All of Us is an NIH initiative that aims to collect genetic, environmental, and health data from one million volunteers from all over the country, with the goal of advancing precision medicine and health equity.

TYPE 2 DIABETES DISPARITIES

Characteristic	Full cohort (n = 86,488) (100%)	Asian (n = 2311) (2.67%)	Black (n = 16,282) (18.83%)	Hispanic (n = 16,966) (19.62%)	White (n = 50,929) (58.89%)
Mean age—years (SD)	55.31 (16.78)	47.70 (16.41)	52.93 (14.70)	48.20 (15.99)	58.78 (16.69)
Female (%)	56,030 (64.78)	1471 (63.65)	10,224 (62.79)	12,155 (71.64)	32,180 (63.19)
Male (%)	30,458 (35.22)	840 (36.35)	6,058 (37.21)	4811 (28.36)	18,749 (36.81)
T2D Prev. (%)	17.21	15.14	21.87	19.92	14.80

Table 1. *All of Us* Study Cohort

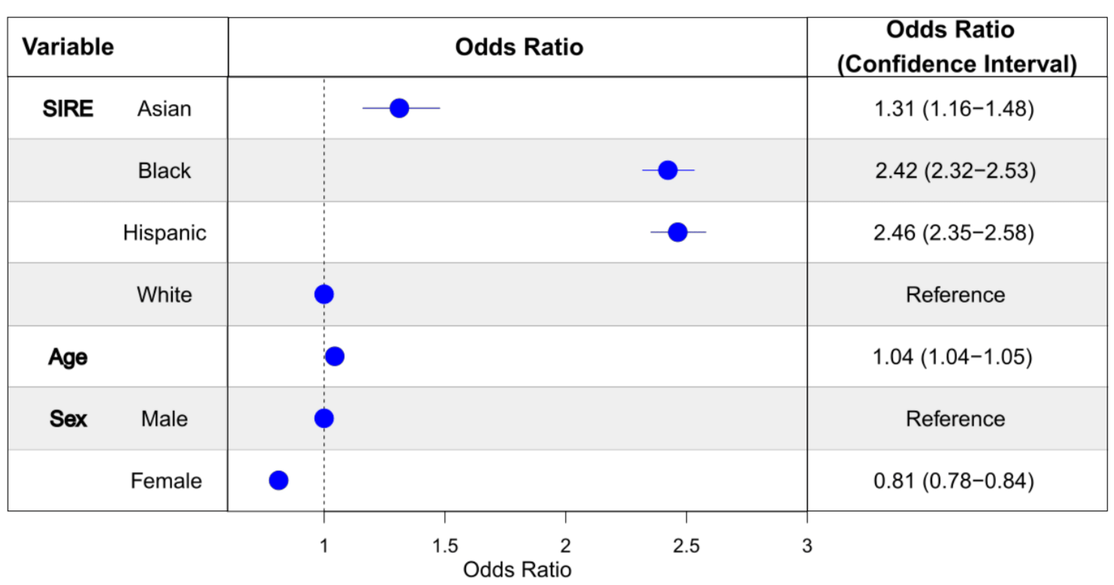

Figure 1. Logistic regression model for the effect of self-identified race and ethnicity (SIRE), age, and sex on T2D outcomes. Odds ratios and 95% confidence intervals are shown for each predictor.

GENETIC ANCESTRY

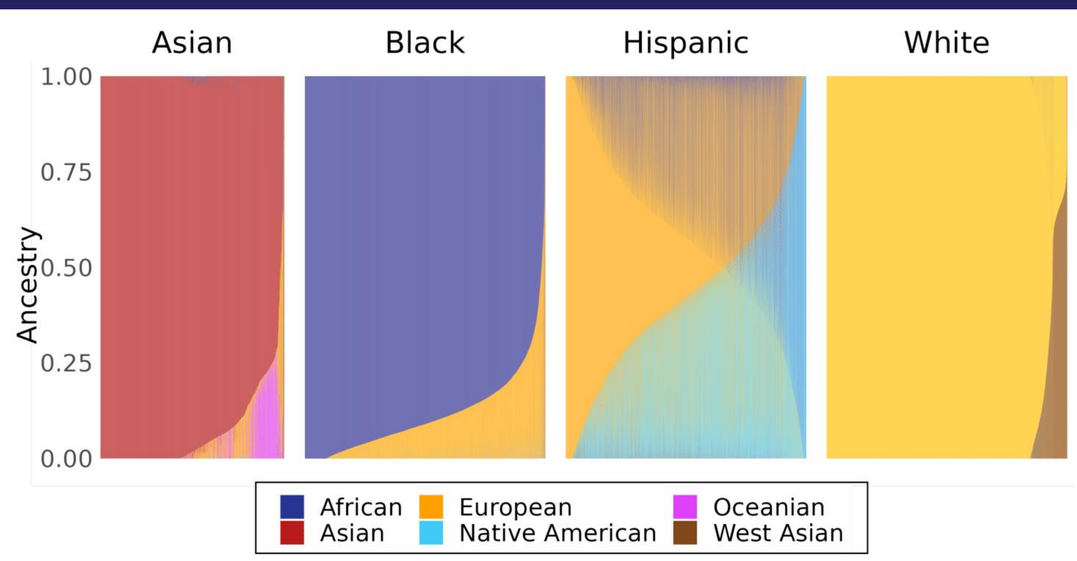

Figure 2. All of Us participant continental genetic ancestry fractions, stratified by SIRE.

SOCIAL DETERMINANTS OF HEALTH

Individual-level socioeconomic deprivation index (iSDI)	Zip code-level socioeconomic deprivation index (zSDI)
Composite metrics made from PCA of constituent indicators of socioeconomic deprivation	
- Individual household income - Employment status - Highest grade of education completed - Home ownership status - Health insurance	- Median area household income - Fraction of population in poverty - Fraction of population with high school diploma - Fraction of population with no insurance - Fraction of population on assisted income - Fraction of houses that are vacant

Table 2. Composite metrics used to measure individual (iSDI) and area-level (zSDI) socioeconomic deprivation (SED). Each metric ranges from 0 to 1, with higher values representing higher levels of SED. iSDI was developed from *All of Us* data by Gupta et al. and zSDI was developed from American Community Survey data by Brokamp *et al*.

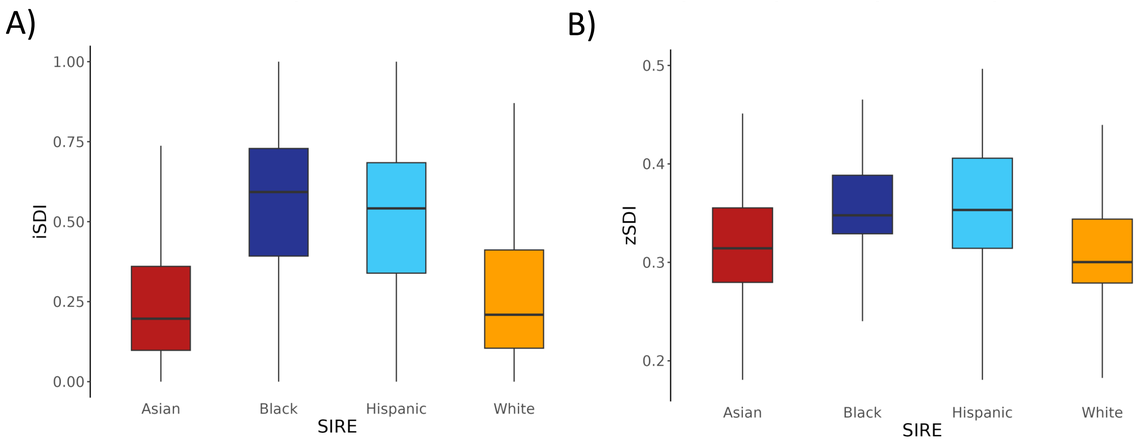

Figure 3. Distributions of iSDI and zSDI values for each SIRE group

MODELING TYPE 2 DIABETES

	Unadjusted		iSDI adjusted		zSDI adjusted		iSDI + zSDI adjusted	
	Estimate (se)	p value	Estimate (se)	p value	Estimate (se)	p value	Estimate (se)	z/t value (p)
Asian	0.04 (0.05)	4.90e-1	0.18 (0.06)	9.74e-4	0.03 (0.05)	5.36e-1	0.18 (0.06)	1.39e-3
Black	0.63 (0.02)	<2e-16	0.27 (0.02)	<2e-16	0.54 (0.02)	<2e-16	0.21 (0.02)	<2e-16
Hispanic	0.70 (0.02)	<2e-16	0.34 (0.02)	<2e-16	0.64 (0.02)	<2e-16	0.30 (0.02)	<2e-16
iSDI			1.60 (0.04)	<2e-16			1.58 (0.04)	<2e-16
zSDI					1.67 (0.12)	<2e-16	1.30 (0.12)	<2e-16

Table 3. T2D risk modeled as a function of varying combinations of iSDI and zSDI.

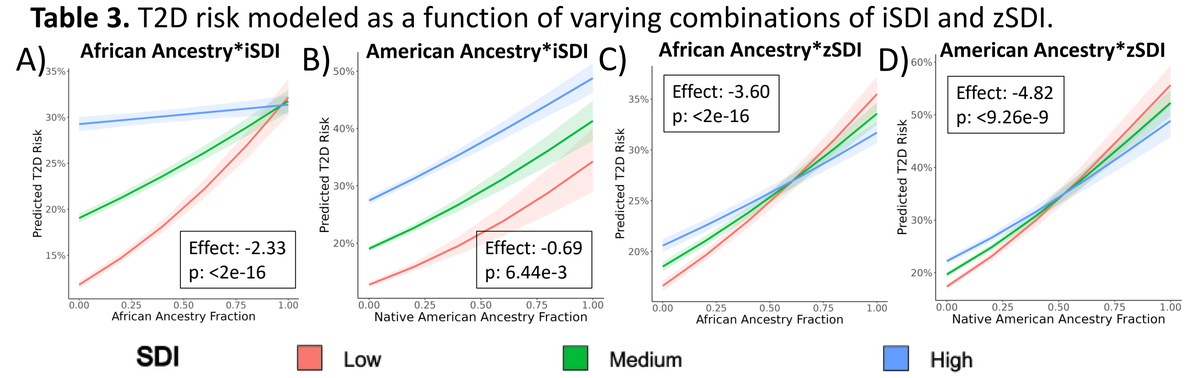

Figure 4. Interaction effects between iSDI (A-B), zSDI (C-D), and ancestry on T2D risk.

SUMMARY

- Racial disparities in T2D can be found in *All of Us*
- Individual and area-level SED contribute to racial disparities in T2D burden in *All of Us*; the contribution of the former appears stronger
- SED is associated with lower T2D risk among those of African and American ancestry

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